

AI/ML intern DAY 5

TASK 2 — ADVANCED ANALYSIS OF AI CLOTHING FITTING ALGORITHMS

LOYA PRANAY (pranay8679@gmail.com)

Date: 15-05-2026

1. Abstract

This task focused on studying Artificial Intelligence-based clothing fitting algorithms used in modern virtual fashion systems. The research explored how AI models analyse body posture, body dimensions, and pose landmarks to digitally align clothes on human bodies.

The study also examined how computer vision and machine learning technologies improve virtual shopping experiences by providing realistic digital clothing simulations.

2. Research Objective

The primary objective of this task was to analyse intelligent garment fitting algorithms and understand how Artificial Intelligence performs virtual clothing alignment on human body structures.

The task focused on:

- digital body analysis
 - body landmark tracking
 - virtual garment simulation
 - clothing alignment systems
 - AI fashion technologies
 - intelligent shopping systems
-

3. Introduction to AI Clothing Fitting Systems

AI clothing fitting systems are advanced computer vision applications that simulate digital clothing on users using body pose estimation and body measurement analysis.

These systems analyse:

- body posture
- body proportions
- body contours
- movement tracking
- clothing dimensions

The AI system then generates a realistic virtual fitting simulation.

These technologies are widely used in:

- online fashion stores
- virtual fitting rooms
- AI shopping platforms
- digital wardrobe systems
- augmented reality shopping applications

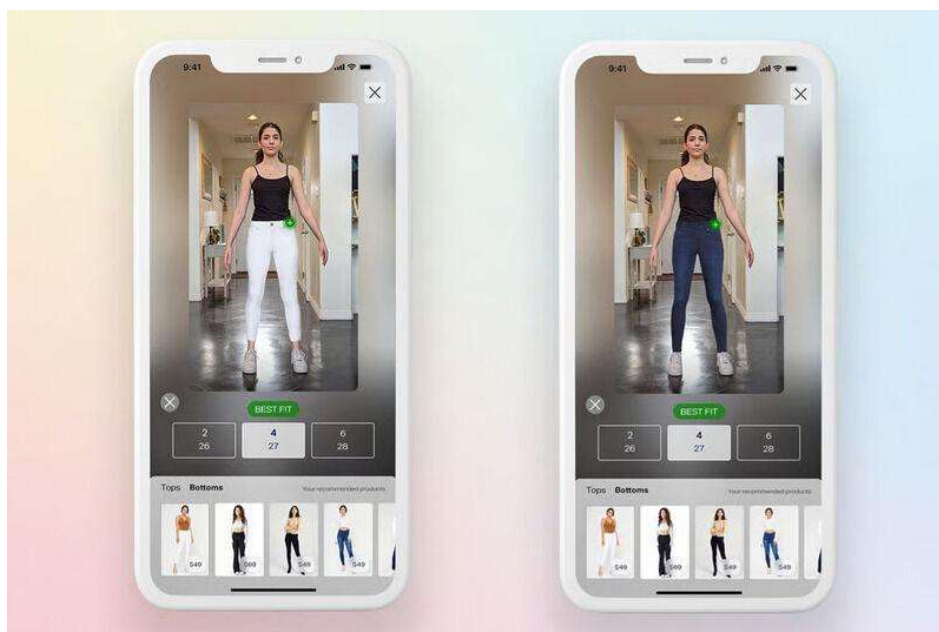


Figure 1 — AI Digital Clothing Fitting

4. Technical Workflow of Clothing Fitting Algorithms

The AI clothing fitting pipeline consists of several processing stages.

Stage 1 — Human Body Detection

Computer vision algorithms first detect the human body from webcam or image input.

The AI system identifies:

- upper body
 - lower body
 - arms
 - legs
 - body orientation
-

Stage 2 — Pose Landmark Extraction

Pose estimation systems extract body landmarks such as:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

These landmarks help estimate body structure and posture.

Stage 3 — Body Dimension Analysis

The AI system calculates:

- shoulder width
- body height
- waist region

- torso proportions

These measurements help estimate appropriate clothing size.

Stage 4 — Garment Alignment

The digital clothing model is aligned according to:

- body shape
 - posture
 - movement direction
 - landmark positioning
-

Stage 5 — Real-Time Visualization

The final clothing simulation is rendered dynamically for real-time viewing.



Figure 2 — AI Body Landmark Mapping

5. Core Technologies Used

Technology	Function
Artificial Intelligence	Smart fitting analysis
Machine Learning	Pattern learning
Computer Vision	Body tracking
OpenCV	Image processing
MediaPipe	Pose landmark detection
Deep Learning	Clothing prediction

6. AI-Based Garment Fitting Techniques

Different AI techniques are used in virtual fitting systems.

Pose Estimation

Used to detect body landmarks and body posture.

Body Segmentation

Separates human body regions from background.

Cloth Simulation

Simulates digital garment movement and fitting.

Landmark Alignment

Aligns virtual clothing according to body coordinates.



Figure 3 — Virtual Garment Simulation

7. Applications of AI Clothing Fitting

AI clothing fitting systems are used in:

- virtual fitting rooms
- e-commerce platforms
- smart retail systems
- personalized shopping
- AI fashion assistants
- digital fashion design

8. Advantages of AI Clothing Fitting Systems

The advantages include:

- reduced product returns
- accurate clothing visualization
- personalized shopping experience
- improved customer satisfaction

- faster clothing selection
- intelligent size estimation

9. Technical Challenges

Challenge	Impact
Poor Lighting	Reduces body visibility
Loose Clothing	Affects body estimation
Fast Body Movement	Reduces alignment stability
Complex Background	Affects segmentation
Incorrect Landmark Detection	Reduces fitting accuracy

10. Comparison of Traditional vs AI-Based Fitting

Traditional Fitting	AI-Based Fitting
Physical fitting required	Digital fitting possible
Time consuming	Faster process
Limited visualization	Realistic preview
Manual size selection	AI-assisted size estimation
Store visit required	Online fitting possible

11. Pseudo Code — AI Clothing Fitting Workflow

START

Capture image or webcam frame

Detect human body using Computer Vision

Extract body landmarks:
shoulders

- elbows
- hips
- knees
- ankles

Estimate body dimensions:

- shoulder width
- body height
- waist region

Load digital clothing model

Align clothing according to:

- body posture
- body position
- landmark coordinates

Adjust clothing size and fitting

Render virtual clothing on body

Display final output

END

12. Algorithm — AI Clothing Fitting System

Step-by-Step Algorithm

Step 1

Initialize webcam and AI pose estimation model.

Step 2

Capture real-time video frame using OpenCV.

Step 3

Convert image from BGR format to RGB format.

Step 4

Detect human body landmarks using MediaPipe Pose.

Step 5

Estimate body measurements using landmark distances.

Step 6

Load virtual clothing image or clothing model.

Step 7

Align clothing according to body posture and landmark positions.

Step 8

Adjust clothing size dynamically based on body proportions.

Step 9

Render clothing overlay on detected body.

Step 10

Display the final virtual fitting result.

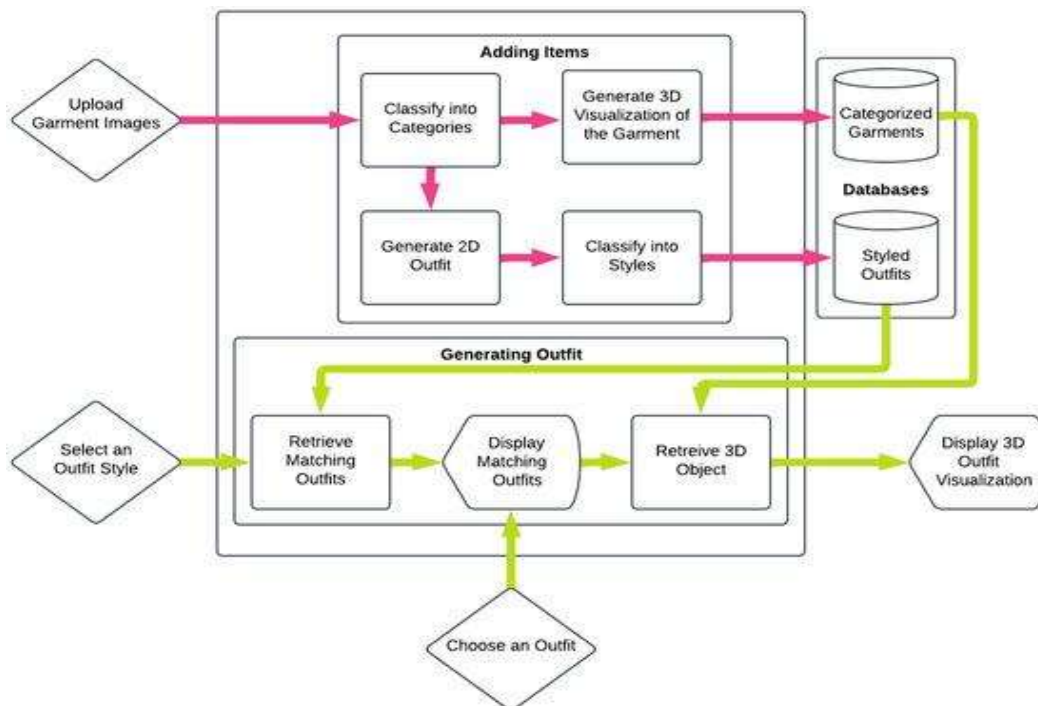
Step 11

Repeat the process continuously for real-time tracking.

Step 12

Terminate the system when exit key is pressed.

Figure 6 — AI Clothing Fitting Workflow



13. Knowledge and Skills Acquired

During this task, the following concepts were understood:

- AI clothing fitting workflows
- body landmark analysis
- garment alignment systems
- virtual fashion technology
- body segmentation
- AI-based pose estimation
- digital shopping systems

14. Conclusion

AI clothing fitting algorithms are transforming modern fashion technology using Artificial Intelligence and computer vision systems.

The research improved understanding of:

- virtual garment simulation
- AI pose estimation
- body landmark tracking
- intelligent shopping systems
- digital fitting technology
- AI fashion applications

These systems are playing a major role in improving online shopping efficiency and personalized customer experiences.